



Riya Rose Roy

✉ riyaroseroy05@gmail.com

☎ +91 8281416541

🌐 [/in/riya-rose-roy](https://www.linkedin.com/in/riya-rose-roy)

🐙 [/Riyaroseroy](https://github.com/Riyaroseroy)

A passionate Electronics and Communication Engineering student shifting analytical hardware precision into software development. Skilled at architecting web applications and scalable backend systems. A collaborative, systems-thinking engineer focused on delivering optimized software solutions. I engineer clean, high-performance digital ecosystems from the database layer to the user interface.

Education

NSS College of Engineering, Palakkad Expected May 2027

B.Tech, Electronics And Communication Engg. — CGPA: **6.98**/10.00

CARMEL HSS CHALAKUDY

March 2023

Higher Secondary Education (Class XII) — Percentage: **98%**

ST. JOSEPH'S EMHS, PAZHOOKARA

March 2021

High School Education (Class X) (School Topper) — Percentage: **97.6%**

Experience

PROJECT INTERN, IIST, Trivandrum (01/05/26 - 30/06/26)

- Developed an autonomous Robotic Arm system using Deep Reinforcement Learning models.
- Optimized reward functions and data structures to minimize path-finding defects and maximize execution accuracy.

Positions of Responsibility

- ENCORE Coordinator, IEEE SIGHT, KERALA SECTION** (2026 - Present): Conducted encore sections across regional campuses.
- Secretary, IEEE SIGHT AG NSSCE** (2024 - Present): Organized the NEXOTECH IEEE Conference 2026 and spearheaded Robocamp Jr.
- IPR LEAD, IEDC NSSCE** (2025 - Present): Coordinated state-level YIP 7.0/8.0 campaigns and represented campus at IEDC Summit 2025.

Projects

AI- Fish Tank

- Engineered an interactive virtual space with real-time gesture tracking and low-latency rendering using deep neural networks.
- Tools Used:** Python, OpenCV, MediaPipe, Pygame, NumPy, gTTS

SHE WANTS - women-friendly platform

- Built a full-stack health web application using a secure, calculator-masked frontend layout to safeguard user privacy.
- Tools Used:** Java, Spring Boot, Python, Django, React, JavaScript, SQL, Google Maps API, JUnit

8-Bit Microprocessor Design

- Simulated an 8-bit microprocessor architecture featuring a modular Control Unit, Register File, and a 24+ op ALU.
- Tools used:** Verilog HDL, Xilinx Vivado, Digital Logic Design, Behavioral Simulation

Smart Automated Delivery Box

- Created an automated box with secure OTP verification, load-cell calibration, and real-time UART-to-cloud updates.
- Tools Used:** Python, Embedded C, UART, Telepot API, IoT Protocols, Hardware-Software Integration

Certifications

- JOY OF COMPUTING PYTHON, NPTEL (Silver Badge) 2025
- DEEP LEARNING: FOUNDATIONS AND CONCEPTS, NPTEL 2026

Achievements

- Competed in national-level hackathons including Bharatiya Antariksh Hackathon 2025 and Cognizant Technoverse.
- Presented the Smart Automated Delivery Box System at the YIP District Evaluation panel.